

The Architecture
of Understanding

The background features a grid of squares on the left, some of which are black. To the right, there is a large, colorful cloud of dots in shades of blue, purple, and pink. A series of black squares are arranged in a curved path from the grid towards the dots, with thin black lines connecting them, suggesting a trajectory or flow.

Tracing the journey from discrete tokens to
continuous meaning in modern language models.

Based on research by Dr Partha Majumdar
(ORC-ID: 0009-0002-7375-8034).

The word once stood as the indivisible atom of meaning

Early foundational models (like Word2Vec and GloVe) operated on a simple assumption: language could be cleanly decomposed into complete words. Each word was a discrete rock, mapped individually into a vector space.



Bounded by a limited vocabulary of 30,000 to 50,000 words.

Early NLP Models



Language is infinite, but computational memory is finite

Treating words as indivisible units creates an impossible bottleneck. Natural language continuously expands with new domains, borrowed phrases, and morphological variations.



The atomic unit is no longer the word, but the token

Modern systems like GPT-4 and BERT do not embed whole words. They utilise advanced algorithms (like Byte Pair Encoding or WordPiece) to iteratively merge frequent character pairs. Rare words are shattered into logical, sub-atomic components.



This hybrid approach expands vocabulary space to a highly efficient 100,000–200,000 tokens.

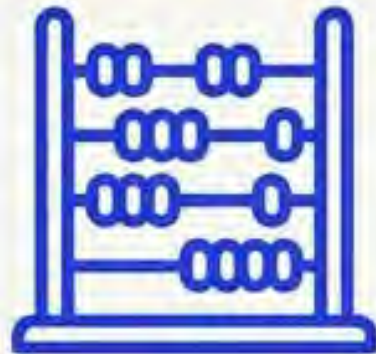
Tokenisation does not assign meaning



It does not learn relationships.



It does not perform inference.



Its sole purpose is **discretisation**:
transforming raw text into a
sequence of **countable indices**.

Three paths to mapping the landscape of language

	Word-Level	Character-Level	Subword (Token)
Granularity	Entire words	Single letters	Reusable components
Vocabulary Size	Massive (Unmanageable)	Tiny (Too small)	Balanced (100k-200k)
Expressiveness	High	Extremely Low	High
Handling “unaffable”	[unaffable] (Fails if unseen)	[u, n, a, f, f, a, b, l, e] (Loses meaning)	[un, ##aff, ##able] (Perfect synthesis)

Balancing vocabulary size with expressive power

Subword discretisation allows common words to remain intact while decomposing complex forms, allowing models to reuse representations across related contexts (e.g., learning, learned, learner).

I am learning language models.



["I"] ["am"] ["learn"] ["ing"] ["language"] ["models"] ["."]

The final output of discretisation is purely numerical

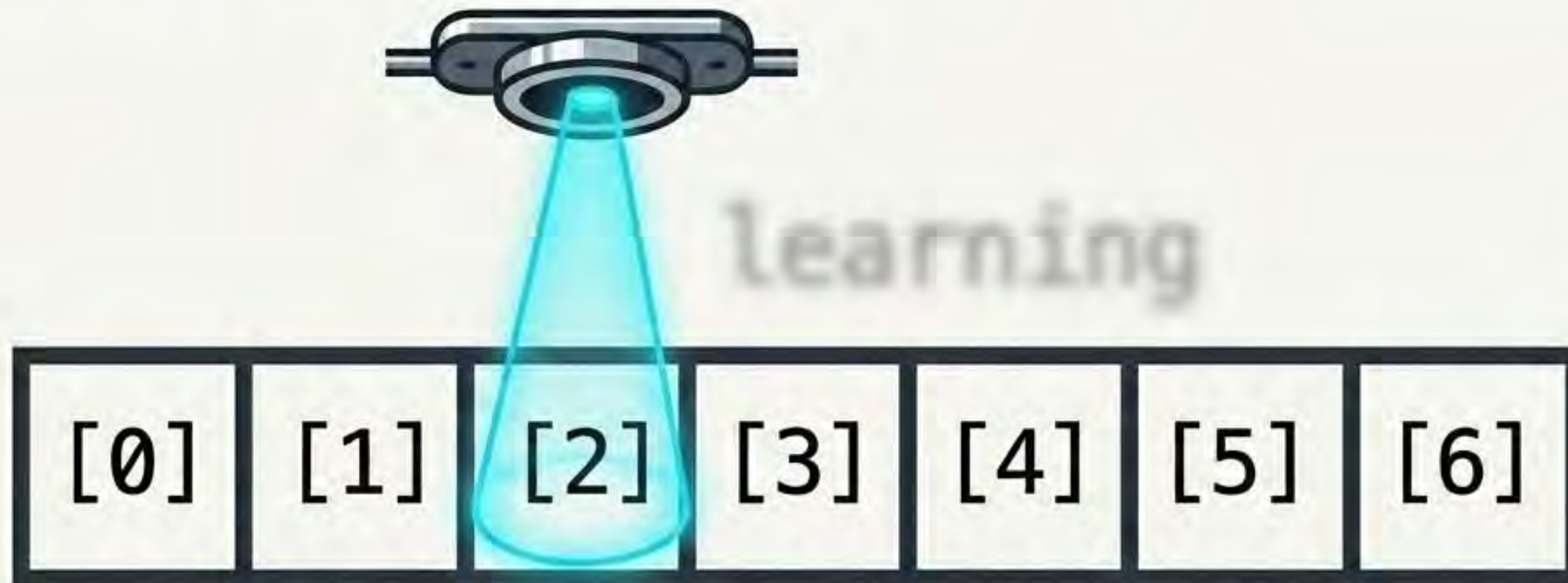
Once a tokenisation scheme is fixed, the text ceases to exist. The sentence is reduced to a sequence of integer identifiers. This is the only form the system receives.



$$t_1, t_2, t_3, \dots, t_n$$

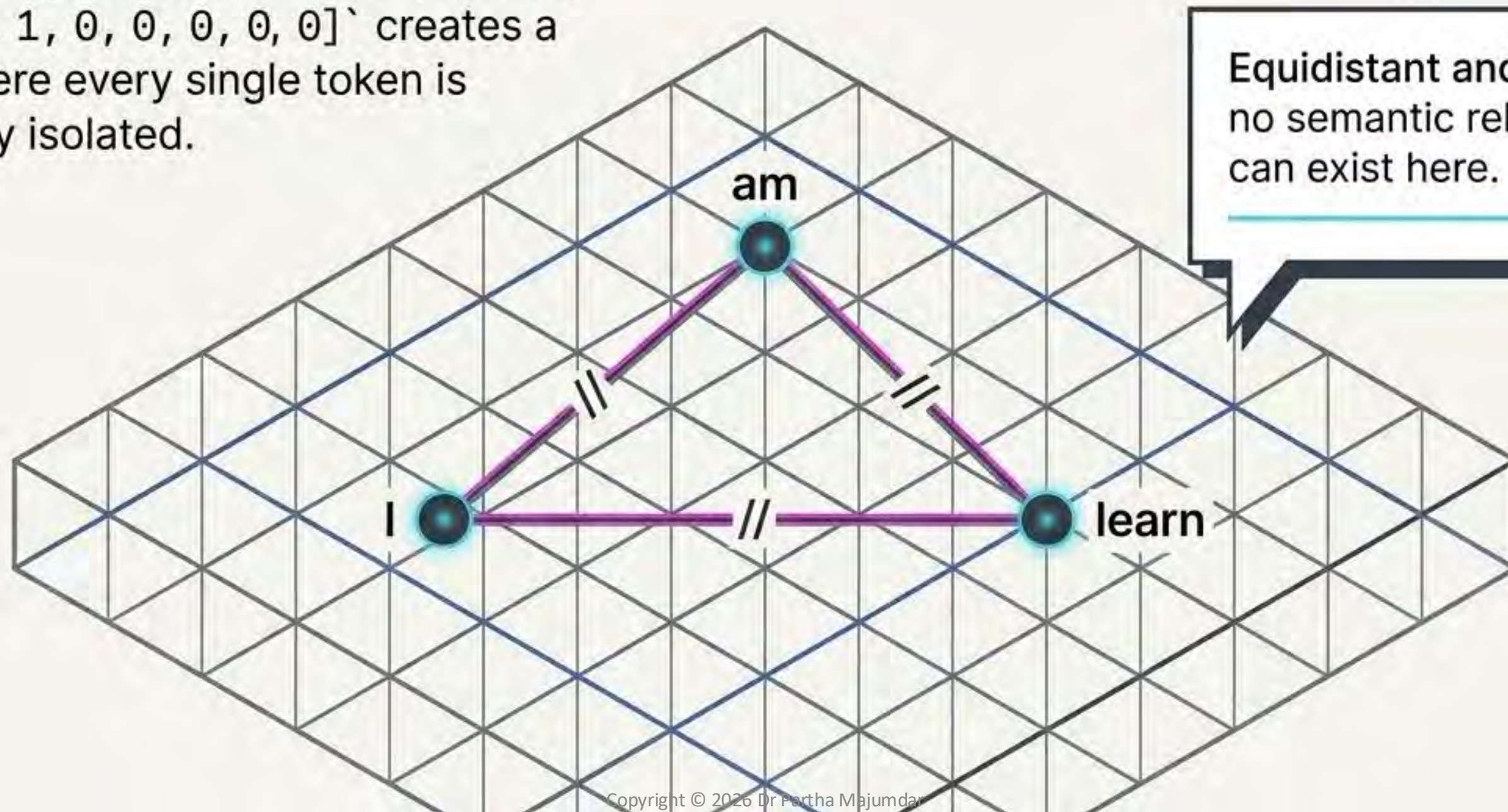
The embedding layer is entirely blind to text

The transition from indices to vectors is the foundational learned step in modern language systems. But at this stage, the indices carry no meaning. They are merely coordinates.



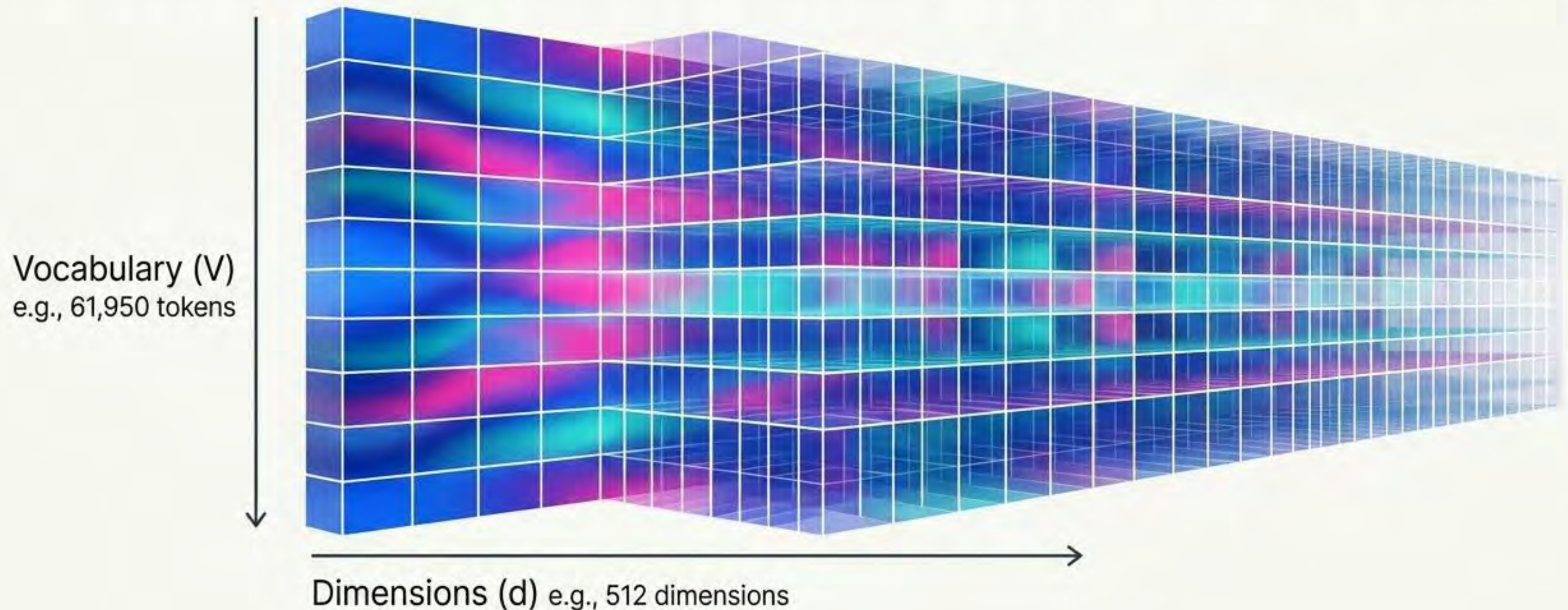
One-hot vectors preserve identity but destroy proximity

Representing the token "learn" (index 2) as `[0, 0, 1, 0, 0, 0, 0, 0]` creates a space where every single token is completely isolated.



The embedding matrix defines the dimensional space

To capture semantic nuance, we introduce $E \in \mathbb{R}^{V \times d}$. This massive, learned lookup table maps a massive vocabulary (V) into a high-dimensional continuous space (d , typically 768 to 2048 dimensions).



Matrix multiplication simplifies to a direct coordinate lookup

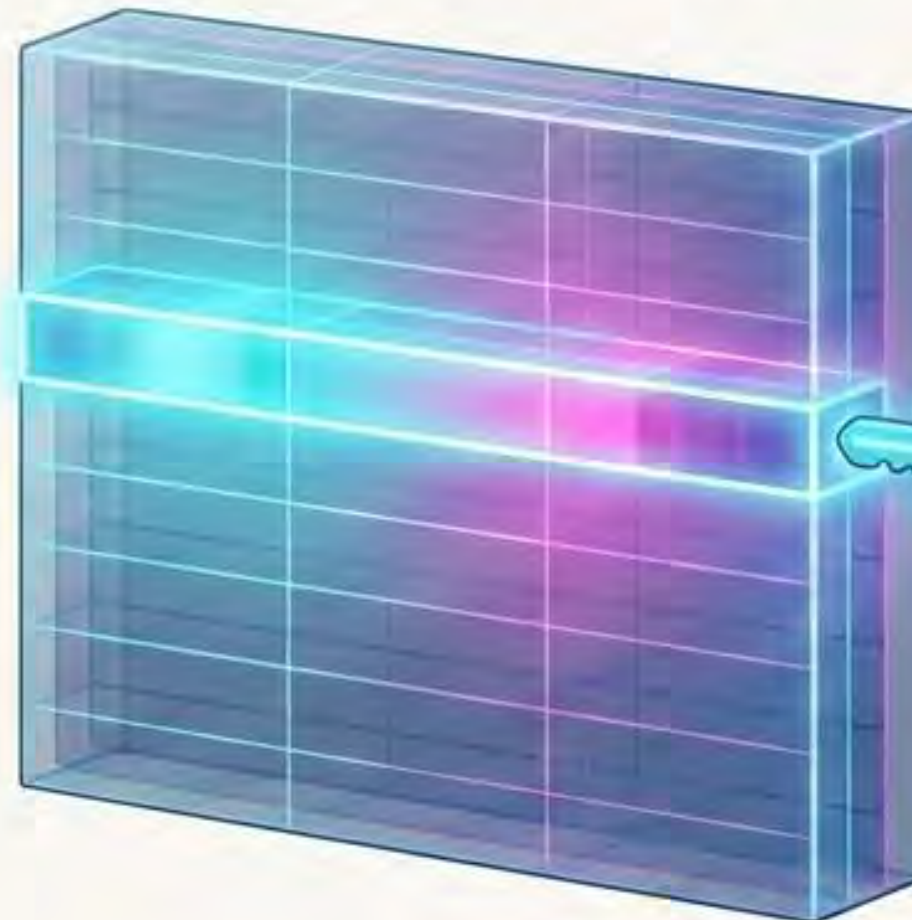
The act of embedding does not require computing a full matrix multiplication. It is mathematically identical to selecting a specific row using the token's index.

One-hot vector array
"learn"

$$\begin{bmatrix} 0, \\ 0, \\ 1 \\ 0, \\ 0, \\ 0, \\ 0 \end{bmatrix}$$



Embedding Matrix E



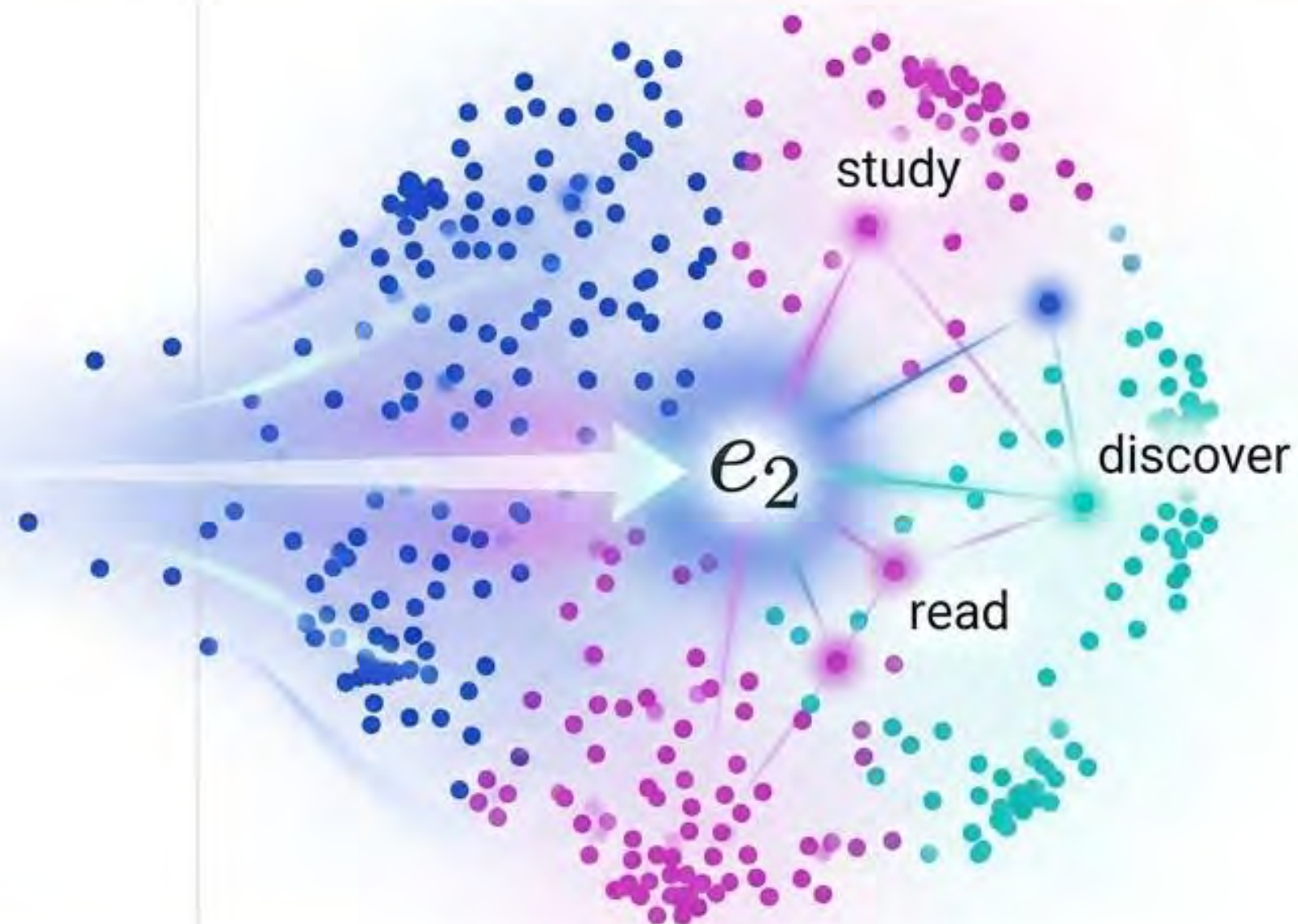
$$e_2 \in \mathbb{R}^4$$



Identifiers are reborn as coordinates of meaning

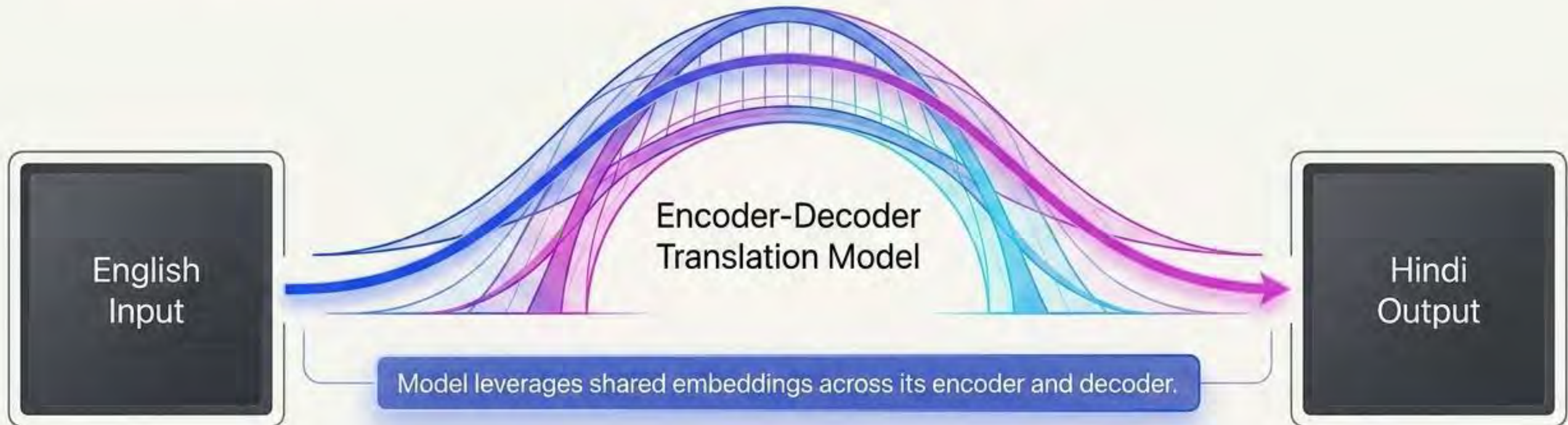
The embedding layer prepares language. It transforms a meaningless integer into a coordinate in a continuous semantic space, ready to be manipulated, compared, and transformed by the network.

[2]



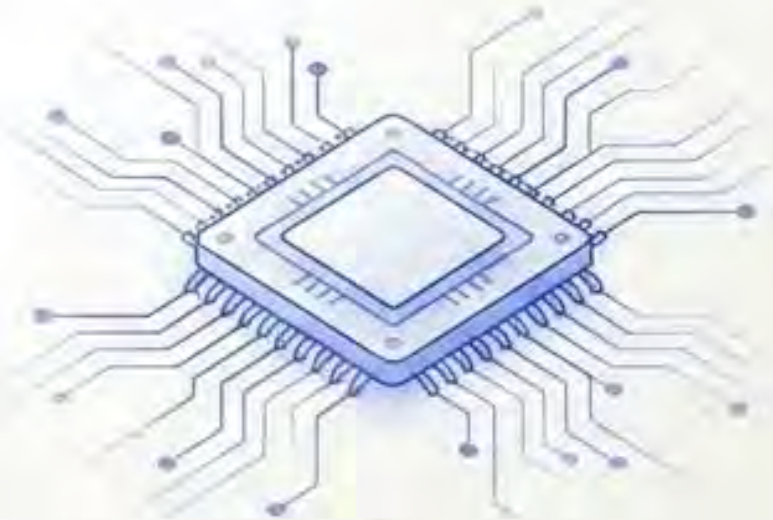
Proving the theory: A cross-lingual translation engine

To observe these concepts in action, we deploy a sequence-to-sequence neural machine translation system: `Helsinki-NLP/opus-mt-en-hi`.



An efficient, constrained experimental environment

We fine-tuned the model using a curated subset of the IITB parallel corpus, balancing computational efficiency with linguistic diversity.



Hardware

NVIDIA Tesla T4 GPU
(16 GB Memory, CUDA 12.8)



Ecosystem

Hugging Face
(Transformers 5.0, Datasets 4.0)

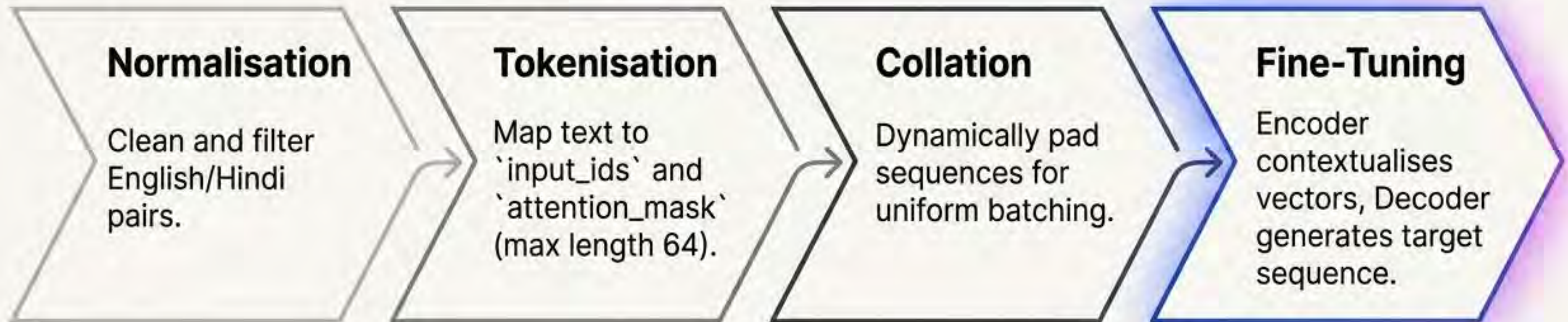


Corpus

46,238 training examples
(Curated subset, reduced from 1.6M for rapid experimentation)

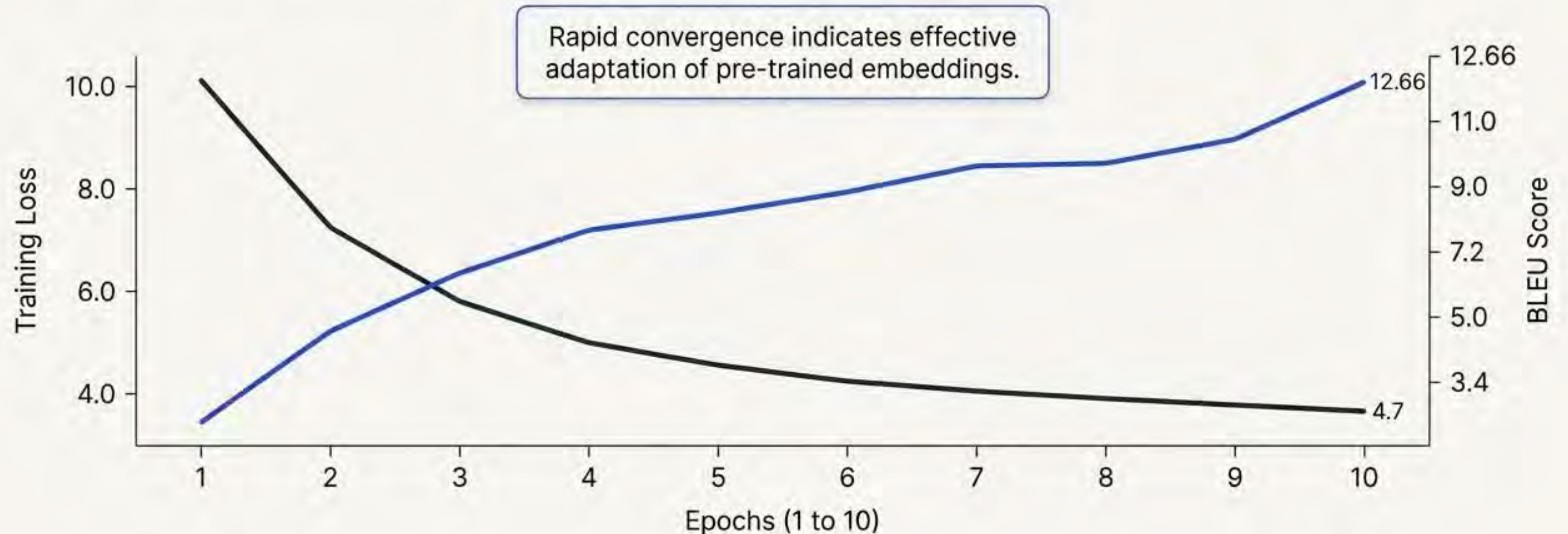
The transformation pipeline: From raw text to trained network

The sequence-to-sequence trainer orchestrates a precise workflow, transforming text into the numerical format required for cross-lingual mapping.



Statistical alignment across languages

The model refines its internal vector representations across 10 epochs. We measure success using the BLEU metric, assessing how closely generated n-grams overlap with human reference translations.



Emergent cross-lingual mapping

Even with a modest BLEU score, qualitative analysis reveals the model successfully captures structural and conversational intent without merely memorising phrases.

English Input		Hindi Output
Hello.		हैलो।
We are studying translation.		हम अनुवाद पढ़ रहे हैं।
This book is about word embeddings.		इस पुस्तक में इस शब्द के बारे में चर्चा की गई है।

Syntax is structurally encoded in the vector space

The model's handling of gender agreement proves that it is applying learned grammatical patterns embedded in its internal representations, rather than performing simple dictionary lookups.



Inspecting the foundational cartography

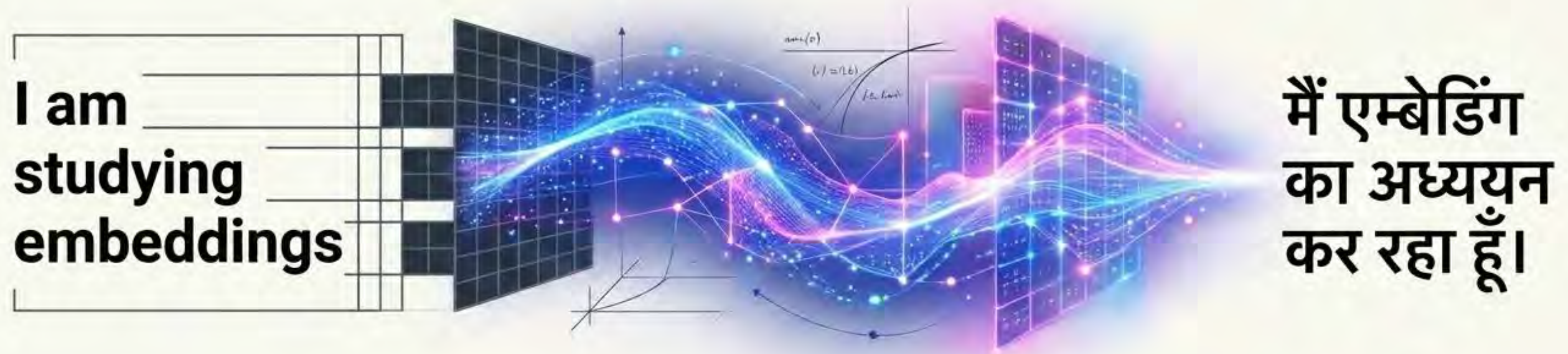
The shared embedding matrix maps over 61,000 tokens to 512-dimensional vectors. This single space bridges both languages.

[61950, 512]

1	39915 (hello)	$[-0.0269, 0.0246, -0.0268, 0.0170 \dots]$
2	5753 (translation)	$[-0.0429, 0.0566, -0.0146, -0.0239 \dots]$

Meaning as an emergent property of the vector space

When the model translates 'I am studying embeddings' into 'मैं एम्बेडिंग का अध्ययन कर रहा हूँ', it is not retrieving a memorised string. It is navigating a shared, continuous space where even highly technical, abstract concepts are situated coherently.



The journey from discrete language to continuous understanding.

Support Independent Research: Fueling Discovery

The Financial Barriers to Research



Essential Infrastructure Costs

Funds are required for software licenses, platform access, and database subscriptions.



Publication & Quality Assurance

Contributions cover article processing charges, professional editing tools, and private reviewers.



How to Support the Work



Discretionary Contributions

Support is entirely voluntary and based on the value found in the sessions.



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Donations can be made directly via the provided PayPal payment link.

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